

Principle Intelligence and Open Structural Learning

From Sparse Differential Evidence to Open-LHS Principles, Two-Way Validation, and Portable Intelligence

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A structural-learning framework in which intelligence can extract reusable Principles from differences, test them against counter-evidence, create or acquire missing structures for reasoning, and externalize validated structures for portable and collective reuse.

Core Research Flow

1 Experience / Observation	2 Difference	3 Candidate Principle	4 Two-Way Validation	5 Knowledge Root	6 Open-LHS	7 PIRP / PIRU	8 Collective Learning
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Four Core Propositions

P1 - Small-Sample Structural Leverage. As evidence becomes sparse, explicit differences, constraints, counter-evidence, and reusable structural invariants can become increasingly valuable.

P2 - Principle Extraction. A Principle is a reusable structural invariant, constraint, or relation extracted from evidence and intended to generalize beyond the observations that produced it.

P3 - Open-LHS Principle. The left-hand side of a Principle is not restricted to a permanently predefined symbolic vocabulary. Existing, retrieved, composed, or runtime-generated intelligence structures may participate in Principle formation and evaluation.

P4 - Principle Lifecycle. A Principle is evidence-bound, context-bound, falsifiable, and evolvable. Counter-evidence may promote specialization, split, merge, weakening, replacement, rejection, or reopening.

Open-LHS and Runtime Structural Generation

LHS vocabulary itself can grow. If reasoning requires a missing structural role, the runtime may search, retrieve, compose, generate, unfold, or delegate a compatible structure, validate it, and bind it into active reasoning.

Key proposition: Reasoning can create or acquire the representational structures required for its own reasoning.

Principle -> Runtime -> Evidence -> Growth

Principle -> Context Binding -> CCC. A Principle preserves reusable structure; a CCC expresses that structure operationally in a specific context.

Two-Way CCC. Runtime consequences are searched back toward structural causes. Supporting evidence and counter-evidence become inputs to Principle revision and Fold Back.

Knowledge Root. A mature Principle can generate CCCs, tests, branches, evaluators, and additional intelligence structures rather than remaining a passive record.

PIRP / PIRU. Explicit intelligence structures can be externalized as Portable Intelligence Runtime Pieces or Units when identity, semantics, interface, evidence, context requirements, and lifecycle are sufficiently explicit.

Collective Structural Learning. Portable intelligence becomes cumulative learning when remote use produces new evidence and that evidence Folds Back into shared structural revision.

Three Canonical Cases

CASE-001 - Direction Matters. A -> B and B -> A may have different behavioral costs; direction itself can carry intelligence.

CASE-002 - Perspective Matters. Visual proximity and behavioral reachability may disagree; the relevant metric depends on task and dynamics.

CASE-003 - Vocabulary Itself Can Grow. Agent A creates PIRU-X; Agent B, without prior training on X, recognizes its structural interface, validates it, binds it into an Open-LHS, uses it, and returns new evidence.

v1.0.0 Research Package

8 Core Papers Rules -> Difference -> Principle -> Open-LHS -> CCC -> Lifecycle -> PIRP/PIRU -> Growth	3 Canonical Cases Direction -> Perspective -> Runtime Vocabulary Growth	7 Architecture Figures Grand Map -> Extraction -> Open-LHS -> Validation -> Lifecycle -> Collective Learning
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Research thesis: Reusable intelligence structures can sometimes be extracted from experience, preserved as evidence-bound Principles, challenged by counter-evidence, revised through structural Delta, admitted into an extensible runtime vocabulary, externalized for portable reuse, and allowed to participate in cumulative structural learning.

Do not merely compute inside the structures you already have. Learn which structures deserve to exist next.

Repository: github.com/sizhet/Principle-Intelligence-and-Open-Structural-Learning